

FINDINGS AND METHOD

ALIGNNN and the PINK Pipeline

A second architecture, the full 554,054-material screen,
and what changed when every claim was checked



PART ONE

Where the project stands

PINK reproduced, then extended – and what the extension is for

What PINK does, in one slide

PINK (Liu *et al.*, *J. Mater. Inf.* 2025) predicts **lattice thermal conductivity** κ_L straight from a CIF file, without solving the phonon Boltzmann transport equation.

It does that in two stages. A crystal graph network predicts the two elastic moduli; a closed-form physical model turns those into κ_L . The physics stage has no free parameters – it is the Slack model, not a fit.

Why this is worth doing

A full DFT + PBTE calculation of κ_L takes days per material. This takes about a sixth of a second. That is the difference between checking a handful of candidates and screening half a million.

The two stages:

Stage 1 – learned

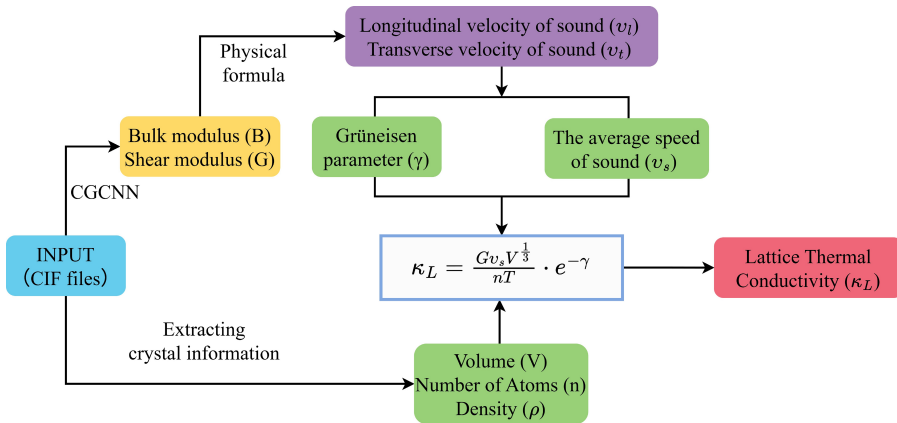
CIF $\rightarrow$ graph network $\rightarrow$ bulk modulus K , shear modulus G



Stage 2 – physics

$K, G \rightarrow$ Poisson ratio $\rightarrow$ sound velocity, Debye temperature, Grüneisen $\gamma \rightarrow \kappa_L$

The paper's own Figure 1



Reproduced from Liu *et al.*, *J. Mater. Inf.* 2025, 5, 12 (CC BY 4.0). **Blue** the input CIF; **yellow** the only learned step; **purple and green** the closed-form physics; **red** the answer. The boxed equation is the one this project reports – note $\kappa_L \propto e^{-\gamma}$, an *exponential* dependence on the Grüneisen parameter.

What this project added

A second architecture

ALIGNN, on the identical split, so the two compare without caveats. It beats the CGCNN ensemble on both moduli.

The full screen, twice

All 33,118 GNoME candidates scored by both architectures – the paper's headline experiment, rerun.

Validation against measurement

45 materials with experimentally measured κ_L , not model-vs-model.

And – the part that took longest –

every headline claim re-checked against the artefacts that produced it. Four evaluation failure modes were found, each of which inflates reported performance. Three numbers already written down turned out to be wrong.

Parts 6–7 are those checks. Part 8 recomputes every stored metric from scratch (0 mismatches); Part 9 is the 15-crystal list now going to DFT.

Two architectures agreeing is evidence. Two architectures agreeing because they share a mistake is not – so most of the work went into telling those apart.

PART TWO

ALIGNN: from CIF to two graphs

The one structural idea that separates it from CGCNN

Why a second architecture at all

A single model that scores well can be right for the wrong reason. Two models that share only the training data and the physics – not the architecture – give an independent check: where they agree, the signal is unlikely to be an artefact of either one.

ALIGNN was chosen because its difference from CGCNN is **physically motivated rather than incidental**. It is not a bigger CGCNN or a differently tuned one. It sees something CGCNN structurally cannot: **bond angles**.

What CGCNN encodes

Which atoms exist, which are bonded, and *how far apart* they are.
A pure two-body description.

What ALIGNN adds

The *angles* between bonds – a three-body term. Elastic response to shear depends on how bonds resist *bending*, not only stretching.

Shear modulus is a bond-bending property. A model blind to angles is being asked to infer it indirectly.

Step 1 — the CIF, and finding neighbours

Identical to CGCNN's start: a CIF gives lattice vectors, fractional coordinates and the element at each site. Neighbours come from a **periodic distance search**, so an atom near a face still finds those across the boundary.

The two settings that define the graph

`cutoff = 5.0 Å` – the radius within which a neighbour counts

`max_neighbors = 12` – the cap per atom

Read from the trained model's own config – building 8.0 Å graphs for a 5.0 Å model would feed it an unseen neighbourhood.

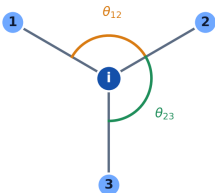
What comes out of this step:

quantity	becomes
element at site i	node feature $\mathbf{v}_i$
bond $i \rightarrow j$	edge $\mathbf{e}_{ij}$
distance r_{ij}	edge feature
angle θ_{jik}	ALIGNN only

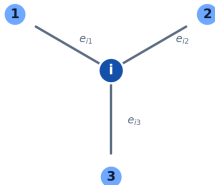
Node features come from the same 92-dimensional element table CGCNN uses (`atom_features = cgcnn` in the config), so the two architectures start from identical chemistry.

Step 2 — one crystal becomes *two* graphs

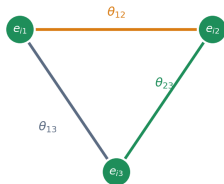
A the crystal
atoms, bonds, and the angles between them



B the ATOM graph $\mathcal{G}$
nodes = atoms, edges = bonds (CGCNN stops here)



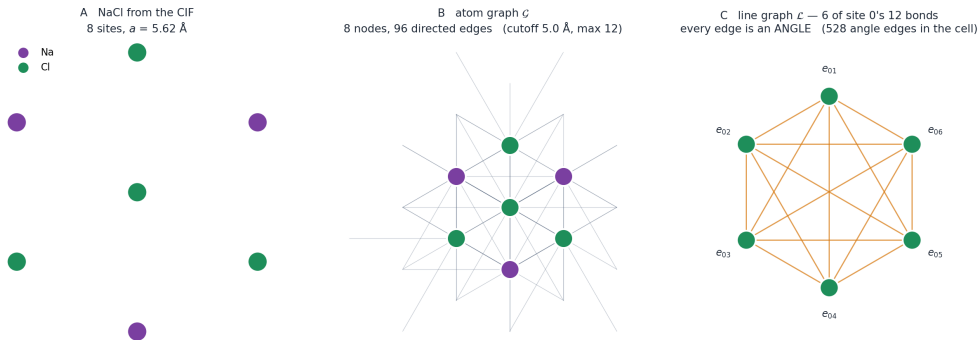
C the LINE graph $\mathcal{L}$
nodes = bonds, edges = ANGLES (ALIGNN adds this)



A the real motif: a centre atom, its neighbours, and the angles between the bonds. **B** the **atom graph** $\mathcal{G}$ – atoms are nodes, bonds are edges. This is exactly what CGCNN builds, and where CGCNN stops. **C** the **line graph** $\mathcal{L}$ – every *bond* of $\mathcal{G}$ becomes a *node*, and two bonds sharing an atom are joined by an edge that carries the angle between them.

The line graph is a graph of the bonds; its edges are angles.

The same thing on a real file: NaCl



Computed from `nacl.cif` at **ALIGNN's own** settings – cutoff $5.0 \text{ \AA}$, max 12 neighbours, from the trained model's config.

A **bond is two things at once**: an edge of $\mathcal{G}$, a node of $\mathcal{L}$.

	node	edge	edge feature
$\mathcal{G}$	atom (8)	bond (96)	length r_{ij}
$\mathcal{L}$	bond (96)	angle (528)	angle θ_{jik}

12 nbrs $\rightarrow \binom{12}{2} = 66$ angles, $8 \times 66 = 528$, so

$528/96 = 5.5 \times$ the edges. Panel **C** shows site 0 only.

Why the line graph is the right construction

A message-passing network updates a node from its neighbours. On the atom graph alone, the only geometric quantity available to a message is the bond *length* r_{ij} – there is nowhere for an angle to live, because an angle is a property of a *pair* of bonds, not of one.

Promoting bonds to nodes solves that exactly. In $\mathcal{L}$, a pair of bonds is an *edge*, and an edge is precisely where a feature can be attached.

Counting the geometry

An atom with n neighbours contributes n nodes to $\mathcal{L}$ and $\binom{n}{2}$ angle edges. At `max_neighbors = 12` that is up to 66 angles per atom – the reason ALIGNN

Angle encoding:

The angle is fed as $\cos \theta_{jik}$ expanded on a radial basis, the same way distances are. Using the cosine rather than the angle keeps the feature smooth where it matters and avoids a discontinuity at 0 and π .

A sanity check

A cubic cell and a sheared one with identical bond lengths are *indistinguishable* to a two-body descriptor, and distinct to ALIGNN.

Step 3 — the ALIGNN layer alternates between them

One ALIGNN layer is two coupled updates, run in order:

1. Update on the line graph $\mathcal{L}$

Each bond-node is updated from the bonds it shares an atom with, using the *angle* on the connecting edge. Bond features now carry angular information.



2. Update on the atom graph $\mathcal{G}$

Each atom-node is updated from its neighbours using those *angle-aware* bond features as edge inputs.

Then, identically to CGCNN

Pool the atom-node vectors over the cell (mean), pass through a small fully-connected head, and emit one number.

Trained on $\log_{10}$ of the modulus, so the loss is relative – a 10 GPa error on a 20 GPa crystal is penalised far more than on a 300 GPa one.

Two separate models, one per target – as in the CGCNN arm.

CGCNN and ALIGNN, side by side

	CGCNN	ALIGNN
graphs built	atom graph only	atom graph + line graph
geometry seen	bond lengths (two-body)	lengths + angles (three-body)
node features	92-dim element table	same 92-dim table
neighbour rule	12 nearest, within 8 Å	12 nearest, 5 Å cutoff
message passing	on $\mathcal{G}$	alternating $\mathcal{L}$ then $\mathcal{G}$
cost per structure	lower	higher (up to $\binom{12}{2}$ angles/atom)
models trained	3 seeds per target	1 + 3 seeds per target
uncertainty estimate	ensemble spread	ensemble spread
bulk K MAE $\log_{10}$	0.0696 one / 0.0630 ens.	0.0539 one / 0.0534 ens.
shear G MAE $\log_{10}$	0.0836 one / 0.0781 ens.	0.0725 one / 0.0708 ens.

Same 10,987 crystals and 1,648-crystal test set (`split_seed` = 42). ALIGNN “one” = its first run (seed 123); “ens.” = three *further* seeds (42, 1, 2). All re-checked in Part 8.

A single ALIGNN beats the three-model CGCNN ensemble. Ensembling buys CGCNN $\approx 8\%$ over its average member, ALIGNN only $\approx 3\%$: its seeds already agree.

One crystal through CGCNN

stage	what happens	shape out	weights
input	each atom: a fixed 92-number element fingerprint (group, radius, ...)	$N \times 92$	–
	each bond: its length spread over 41 Gaussians, 0–8 Å in 0.2 Å steps	$N \times 12 \times 41$	–
embed	one linear layer, $92 \rightarrow 64$	$N \times 64$	5,952
conv $\times 3$	per neighbour: [self 64 neighbour 64 bond 41] = 169 $\rightarrow$ 128, halves: a <i>gate</i> (sigmoid, 0–1) $\times$ a <i>message</i> (softplus); sum the 12 gated messages, add to the atom's own vector	$N \times 64$	$3 \times 22,144$
pool	average over the N atoms of the cell – size-independent	1×64	–
widen	softplus, linear $64 \rightarrow 128$, softplus = the hidden features	1×128	8,320
output	a weighted sum of those 128 numbers + a bias = $\log_{10} K$ (scaled)	1	129
total			80,833

N = atoms in the cell. The gate is the CGCNN idea: the network learns, per neighbour, *how much* to listen (gate) separately from *what* it hears (message). Three rounds of this let information travel three bonds.

ALIGNN keeps the same skeleton – embed, message-pass, average, one number out – but runs it on two graphs (atoms and bonds-with-angles), 4 + 4 layers, 256 wide: ≈ 4.0 million weights, about $50 \times$ CGCNN.

Training settings, from the saved runs

	CGCNN	ALIGNN
read from	checkpoint .pth	config.json per run
target	$\log_{10} K$ or $\log_{10} G$	same, one model each
split	7,691 / 1,648 / 1,648	same crystal ids
neighbours	12 nearest within 8 Å	12 nearest, 5 Å cutoff
bond input	41 Gaussians	80 (lengths) + 40 (angles)
width	atoms 64, head 128	embed 64, hidden 256
layers	3 graph convolutions	4 ALIGNN + 4 GCN
weights	80,833	≈ 4.0 M
loss	mean squared error	mean squared error
optimiser	Adam, lr 0.01	AdamW, lr 0.001
weight decay	10^{-5}	10^{-5}
schedule	cosine	one-cycle
batch / epochs	128 / 200	64 / 150
seeds	42, 1, 2	123; then 42, 1, 2

Why $\log_{10}$, not GPa

Moduli span more than two orders of magnitude. In log space a 10% miss costs the same on soft and stiff crystals – and the screen lives among the soft ones.

Why two models, not one

K and G each get their own network. The probe on 12c shows why: the K network carries G only as far as K itself does.

Not identical recipes

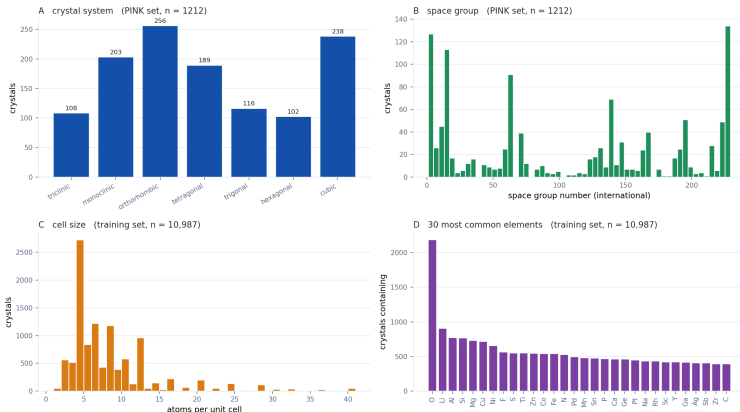
Split, targets, loss and decay match; optimiser, rate, schedule, batch and epochs are each model's own default. The gap is architecture *plus* recipe.

PART THREE

Training and accuracy

The dataset, the fit, and where it fails

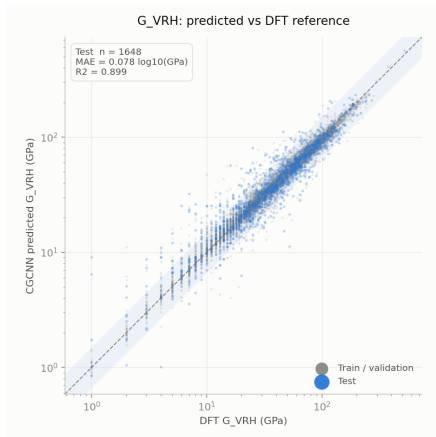
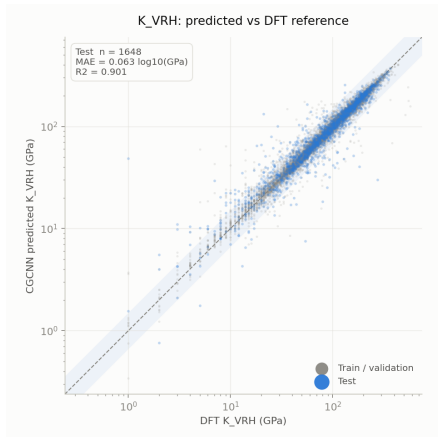
The training set — PINK's Figure 4



matbench: **10,987 crystals**, split 70/15/15 at `split_seed = 42`; both moduli right-skewed, hence $\log_{10}$. **The soft tail in panel B is thin – and it is exactly where the screen operates.**

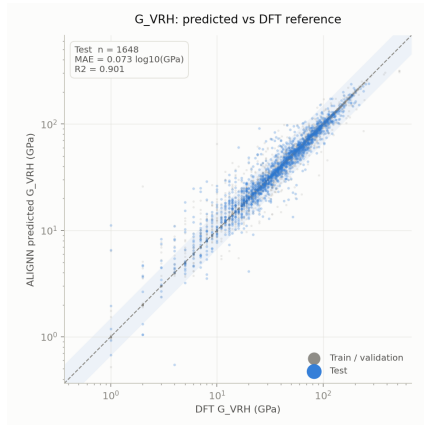
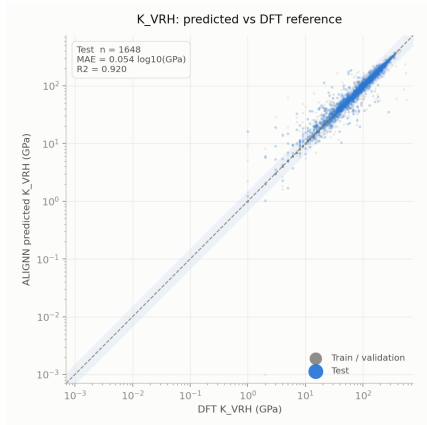
10 • RESULTS

Predicted vs DFT — PINK's Figure 5, CGCNN arm



3-model ensemble, 1,648 held-out crystals. Bulk 0.0630, shear 0.0781 MAE $\log_{10}$.

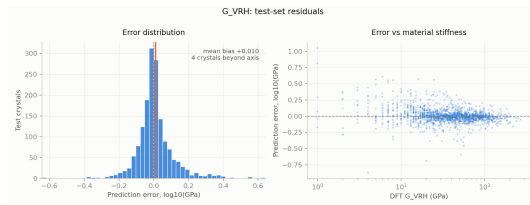
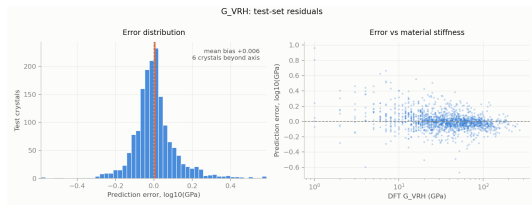
Predicted vs DFT — the ALIGNNN arm



Same crystals and axes. Bulk 0.0539 ($R^2 = 0.920$), shear 0.0725 ($R^2 = 0.901$). **Artefact left visible:** one *negative* bulk modulus, which a log-space model cannot produce.

12 • RESULTS

Where the models fail — residuals



CGCNN ensemble, shear

ALIGNN, shear

Residuals fan out below ~ 30 GPa in both. Stratified by true κ_L , the softest quartile has MAE $\log_{10}(G) = 0.134$ against 0.057 for the stiffest – a factor of 2.4.

Both architectures are worst in exactly the regime the screen targets.
Part 7 asks whether that is the model or the labels.

The hidden features of CGCNN

The last layer of CGCNN (slide 08b) is only a weighted sum:

$$\log_{10} K = w_1 h_1 + w_2 h_2 + \cdots + w_{128} h_{128} + b$$

The 128 numbers h are the **hidden features** – everything the network knows about a crystal when it predicts.

How they were read: a “hook” copies h as each of the 1,648 test crystals passes through the trained K model (seed 42). Nothing is retrained.

How they are drawn: PCA rotates the 128-dimensional cloud so axis 1 follows its largest spread, axis 2 the next; the map plots those two.

Before answering, the network squeezes a crystal to about two numbers. The first is, in effect, “how stiff”; slide 12c shows what the second separates.

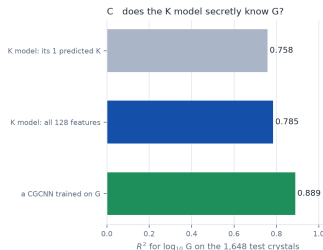
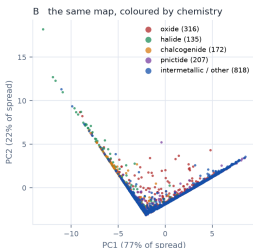
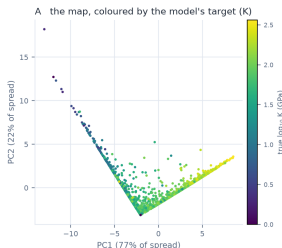
Checked first

This run’s predictions equal the saved ones to 2.4×10^{-7} in $\log_{10}$ – the right model, crystals and graphs were loaded.

128 numbers, effectively two

share of the spread on PC1	76.6%
on PC2	22.4%
on PC3	0.85%
correlation of PC1 with true $\log_{10} K$	+0.91

The hidden-feature map of 1,648 crystals



the 127 extra numbers add only +0.026: G is mostly 'K in disguise'

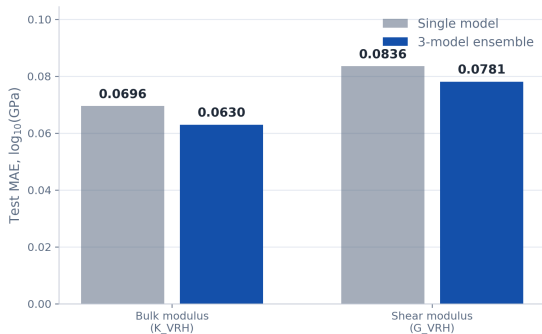
A. Soft (dark) to stiff (yellow) runs along PC1. The map is a **V**: two arms meeting at a tip.

B. Left arm 611 crystals (median K 41 GPa), right arm 1,037 (129 GPa). The **72 off the arms** have 3× the error (median 0.099 vs 0.034 $\log_{10}$); 40 are oxides. Off-arm cut 1.0: set by eye, not tuned.

C. A straight-line probe reads G from the 128 features at $R^2 = 0.785$ – but predicted K alone gives 0.758. Only +0.026 is new; a real G model reaches 0.889.

Where a crystal lands on this map flags where the model is unsure. And the K network knows G only through K – the reason G gets its own network.

What ensembling buys



Three models per target, different random initialisations, averaged in $\log_{10}$ space – a geometric mean, which is the right notion of middle for a quantity whose error is proportional.

Ensembling removed 9-10% of the error:
bulk $0.0696 \rightarrow 0.0630$, shear
 $0.0836 \rightarrow 0.0781$.

And a free uncertainty estimate

The spread across members gives a per-crystal error bar – the thing PINK's single pretrained model cannot produce at all.

Part 4 shows that estimate is badly overconfident.

PART FOUR

The physics

From two moduli to a thermal conductivity

Equations 1–3 — from moduli to sound

The two predicted moduli fix how fast sound travels, and phonon speed sets thermal conductivity.

$$v_l = \sqrt{\frac{K + \frac{4}{3}G}{\rho}} \quad \text{longitudinal wave speed} \quad (1)$$

$$v_t = \sqrt{\frac{G}{\rho}} \quad \text{transverse wave speed} \quad (2)$$

$$v_s = \left[\frac{1}{3} \left(\frac{1}{v_l^3} + \frac{2}{v_t^3} \right) \right]^{-1/3} \quad \text{averaged sound velocity} \quad (3)$$

sym	meaning	unit
K, G	bulk, shear modulus (<i>predicted</i>)	GPa
ρ	mass density (<i>from the CIF</i>)	g cm^{-3}
v_l, v_t, v_s	sound velocities	m s^{-1}

Equations 4–6 — Debye temperature and Grüneisen

$$\Theta_D = \frac{h}{k_B} \left(\frac{3}{4\pi V} \right)^{1/3} v_s \quad \text{Debye temperature} \quad (4)$$

$$\nu = \frac{(v_l/v_t)^2 - 2}{2(v_l/v_t)^2 - 2} \quad \text{Poisson's ratio} \quad (5)$$

$$\gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)} \quad \text{Grüneisen parameter} \quad (6)$$

sym	meaning	unit	sym	meaning	unit
h	Planck const.	J s	Θ_D	Debye temp.	K
k_B	Boltzmann const.	J K ⁻¹	ν	Poisson's ratio	–
V	vol. per atom (CIF)	Å ³	γ	Grüneisen	–

Eq. 6 matters most: γ is derived from the same K , G .

Equations 7–8 — two routes to κ_L

$$\kappa_L^{\text{Slack}} = \frac{A(\gamma) \bar{M} V^{1/3} \Theta_D^3}{\gamma^2 T N}, \qquad A(\gamma) = \frac{2.43 \times 10^{-8}}{1 - \frac{0.514}{\gamma} + \frac{0.228}{\gamma^2}} \tag{7}$$

$$\kappa_L^{\text{cal}} = \frac{G v_s V^{1/3}}{N T} e^{-\gamma} \qquad \text{(the boxed equation in the paper's Figure 1)} \tag{8}$$

sym	meaning	unit	sym	meaning	unit
$\bar{M}$	mean atomic mass	amu	T	temperature, 300	K
N	atoms per cell	–	κ_L	conductivity	W m ⁻¹ K ⁻¹

This project reports κ_L^{cal} , whose γ dependence is *exponential*, not γ^{-2} – so swapping γ rescales it by $e^{\gamma_{\text{old}} - \gamma_{\text{new}}}$.

The Slack chain — no free parameters

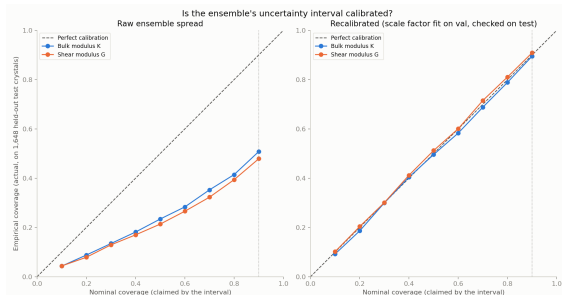


Every arrow is a closed-form relation. The only learned quantities in the whole chain are K and G ; everything downstream is algebra plus the structural constants (volume, density, atom count) read straight from the CIF.

γ is **derived** from the predicted K/G ratio through an empirical Poisson relation – matbench contains no measured γ at all. That single choice turns out to dominate several later results.

Because the physics is shared and unchanged, any difference between two of this project's screens is the *moduli*, never the formula.

The uncertainty interval was badly overconfident



The Monte Carlo p05-p95 band was checked against the test split's real DFT labels – no new data needed.

Claimed 90% coverage. Actual 48-51%.

Cause: three identically-trained members share blind spots, so member *disagreement* runs 4-6× smaller than true error.

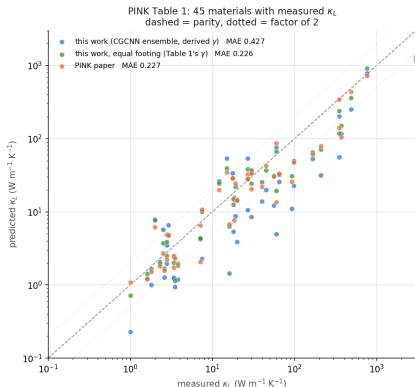
Recalibration factors from validation ($K \ 4.27\times$, $G \ 4.54\times$) recover $\sim 90\%$ coverage out of sample – and collapse almost every “tight” label on the DFT shortlist to “wide”.

PART FIVE

Validation against measurement

45 materials where the answer is known

Measured κ_L — PINK's Figure 6B, our version



The paper's Table 1 lists 46 materials with **experimentally measured** κ_L . 45 survive (mp-3490/GaP is deprecated in Materials Project).

The range spans four orders of magnitude — AgCl at 1.0 to diamond at $3000 \text{ W m}^{-1} \text{K}^{-1}$ — so everything is scored in $\log_{10}$.

mean $|\Delta \log_{10}|$

ours, derived γ 0.427

ours, equal footing **0.226**

the PINK paper 0.227

At face value we look twice as bad; on equal footing we match it.

The comparison was never like-for-like

Table 1's γ values come from **AFLOW / experimental lookups** – the paper's own text says so. They are *not* produced by the Poisson-ratio formula that our pipeline uses.

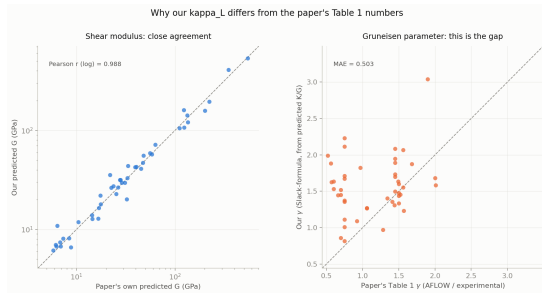
Neither does the paper's own released screening code use them: it uses the same formula we do. So comparing our derived γ against a looked-up- γ table tests **the γ source**, not the model.

Substituting Table 1's own γ

changes nothing else – same moduli, same physics –

0.427 → **0.226**

against the paper's 0.227.



Our moduli were never the problem: shear modulus and sound velocity agree with the paper's own tabulated values at $r \approx 0.99$.

The comparison sheet

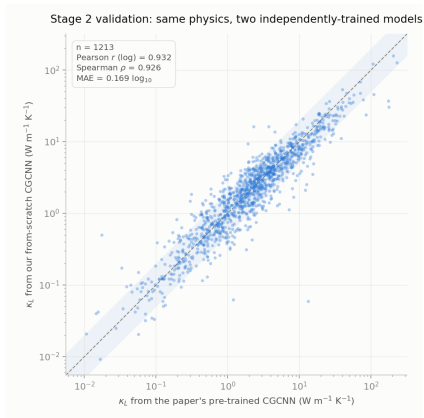
material	measured	ours (derived γ)	ours (equal footing)	PINK
AgCl	1.0	0.23	0.71	1.09
Bi ₂ Te ₃	1.6	1.24	1.43	1.20
NaI	1.8	1.00	1.66	1.52
PbTe	2.5	5.72	3.74	2.71
... 37 more ...				
AlN	350.0	55.87	118.49	140.93
SiC	490.0	251.19	360.10	438.31
BN	760.0	779.67	913.63	727.98
C (diamond)	3000.0	1180.18	1260.63	1320.87

results/cgcnn/63_experimental_vs_predicted.csv – 45 rows, 27 columns. Both γ conventions are carried per material, along with each model's moduli, all three γ values, and the p05/p95 interval, so the choice is visible rather than baked in.

ALIGNN is worse here – 0.450 against the CGCNN ensemble's 0.427.

It wins on matbench moduli and loses on measured κ_L . That was tested, not assumed.

Two architectures, independently, on the same 1,213 crystals



ALIGNN's moduli pushed through the *unchanged* Slack chain, compared against the CGCNN ensemble's κ_L on the same crystals.

Pearson $r = 0.940$, **Spearman** $\rho = 0.947$,
MAE 0.141 $\log_{10}$ – tighter than CGCNN's
own agreement with the paper's pretrained
model ($r = 0.932$).

Two independently trained architectures, sharing only the data, the split and the physics, converging this closely on final κ_L is evidence the pipeline finds physical signal.

PART SIX

The screen

554,054 materials, scored twice

The screening funnel — PINK's Figure 7

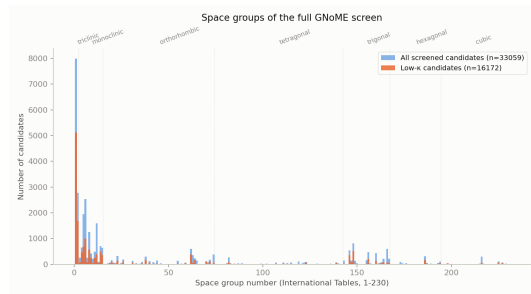
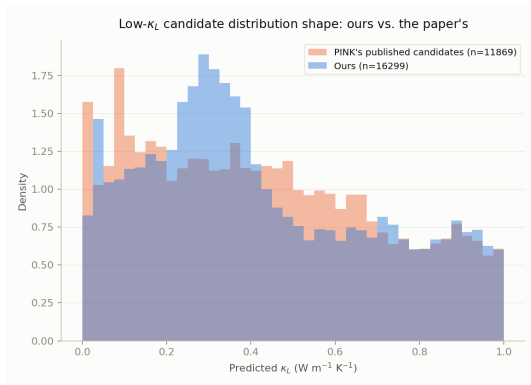


The current GNoME snapshot holds **554,054** materials against the paper's 377,221 – stated explicitly rather than glossed, because it means the candidate lists are not directly comparable.

Against the paper's own published 11,869 candidates, **70.8%** also appear in our independently derived list. Two pipelines sharing only an architecture family and a physics formula, on overlapping-but-different corpora, converging on most of the same discoveries.

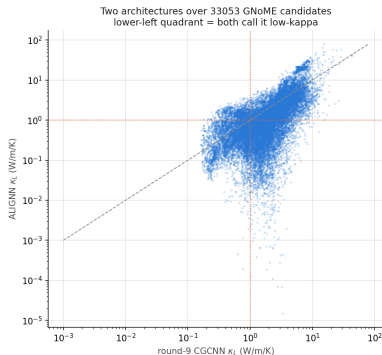
Flagged: that **70.8%** predates several corrections in Part 7 and should be recomputed before it is quoted in a paper.

What the candidates look like — PINK's Figure 8



κ_L distribution and crystal-system breakdown of the screened pool. Halogens are $1.4\text{--}1.9\times$ enriched among low- κ candidates, oxygen $0.27\times$ depleted.

ALIGNN over the whole screen, for the first time



All 33,118 candidates scored with ALIGNN – 68 minutes, zero CIF failures. Only the moduli change; the Slack chain and the structural constants are identical, so nothing downstream can leak a difference.

low- κ call on 33,053 reliable candidates

ALIGNN 14,711 (44.5%)

round 9 7,262 (22.0%)

both agree 6,006

Spearman ρ 0.723

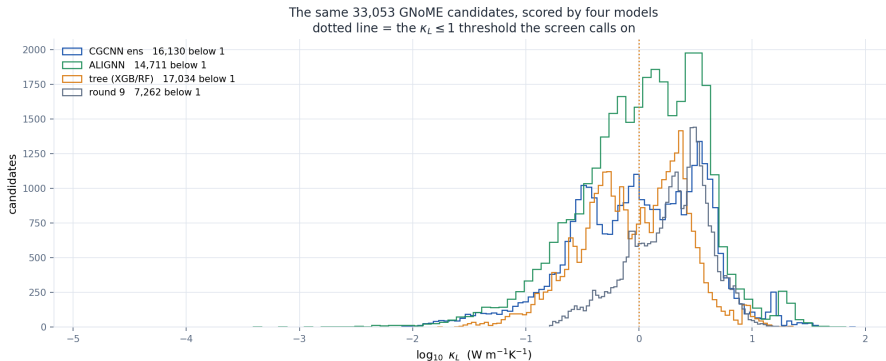
65 hit ALIGNN's $\sim 10^{-3}$ GPa regression floor – flagged, excluded.

PART SEVEN

What we found when we checked

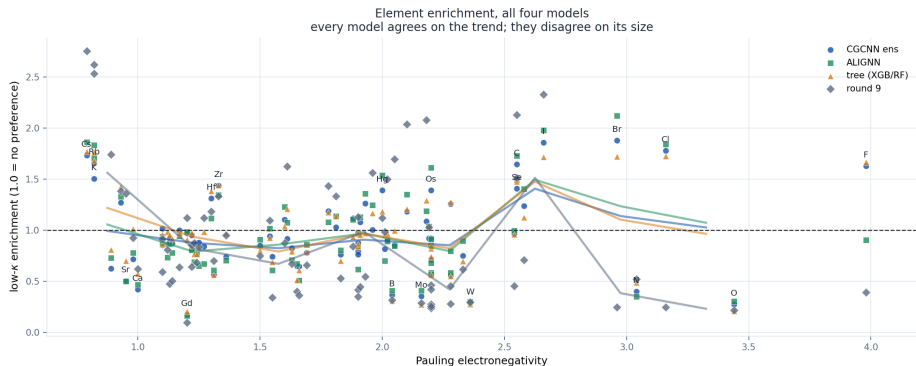
Four evaluation failure modes, and three numbers that were wrong

The same candidates, four different answers



Every model scores the identical 33,053 candidates through the identical physics – only K and G differ. Low- κ counts: **16,130** (CGCNN), **14,711** (ALIGNN), **17,034** (tree), **7,262** (round 9). Three agree closely; round 9 sits far right and calls less than half as many.

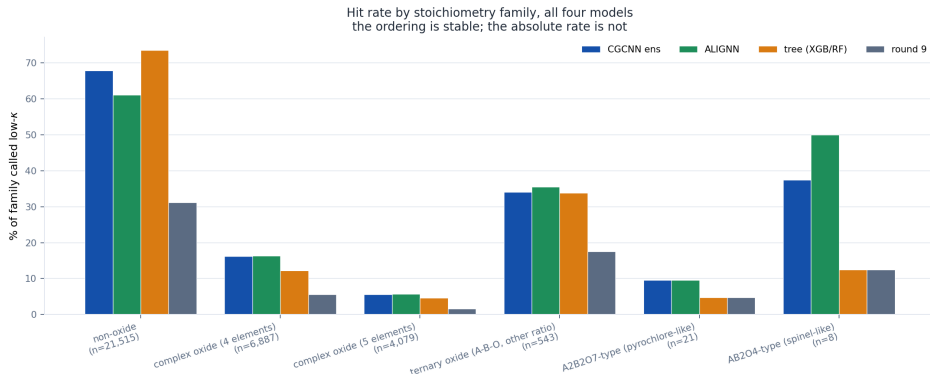
Classifying the candidates: elements



Enrichment is how often candidates containing an element are called low- κ , over that model's own base rate.

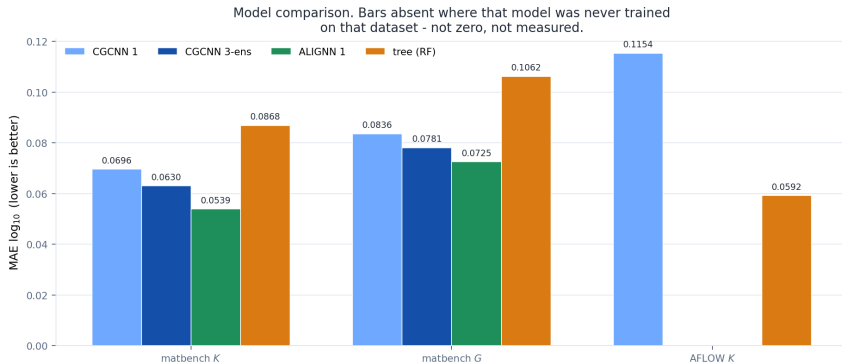
Lines are binned medians. **Cs, Rb, K and the halogens Br, Cl, F sit high; O and N sit lowest** – which Equations 1-3 predict, since a low G gives a low v_s . **All four agree on the shape**; round 9 diverges at the electronegative end.

Classifying the candidates: structural families



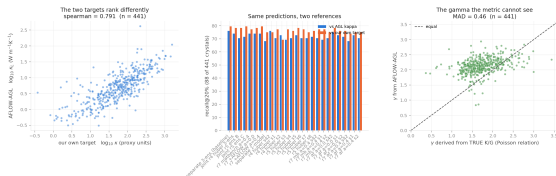
Non-oxides dominate, but *how* much depends on the model: 68% (CGCNN), 61% (ALIGNN), 73% (tree), 31% (round 9). **The family ordering is stable across all four; the absolute rate is not.**

Every model this project has trained



Bars absent where a model was never trained on that dataset – not zero, not measured. **The composition-only tree beats the CGCNN on AFLOW** (0.0592 vs 0.1154) while losing on matbench: underfitting, not a limit of graph networks, since the network's *training* error there (0.103) exceeds the tree's *test* error.

The screening metric was circular



recall@10% – the fraction of the true lowest- κ decile recovered – was scored against a reference that was *this project's* own Slack chain on matbench's true moduli, with γ derived by the same relation the prediction side uses.

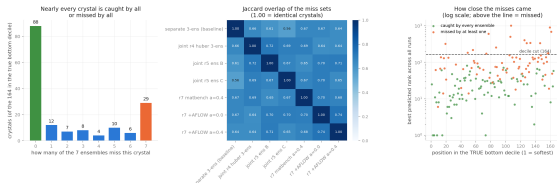
Slack cancels. γ cancels harder: **a model predicting a better γ scored worse.**

Re-scored against AFLOW-AGL's independent κ_L (441 matched crystals): recall@10% fell **20.5 points**, Spearman ρ fell **0.139**.

The old metric actively penalised the one remaining lever. **Quote** Spearman ρ – it is what transfers between targets ($\rho = 0.92$ across 26 runs); recall@10% does not ($\rho = 0.20$).

24 · FINDINGS

The misses are a fixed core, not variance



Fingerprinting the saved prediction vectors collapsed 34 roster entries to **26 distinct** runs – eight were byte-identical duplicates.

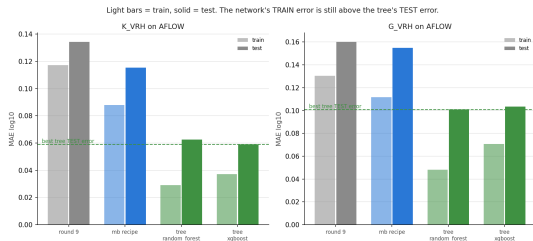
Across those 26, recall spans **64.0-70.7%**. Which crystals are missed is remarkably stable: **16 missed by all 26 runs**, 29 by all seven ensembles, 65 never missed.

The 16 permanent misses: TmAg_3 , MgAgAs , CeTe , $\text{Cs}_2\text{NaAlH}_6$, InSb , YSe and ten others – median 4 sites, 9 of 16 cubic, never ranked better than 173rd by anything.

The ceiling is in the inputs, not the architecture. Seven rounds of architecture work never moved it.

25 • FINDINGS

A composition-only tree beats the network



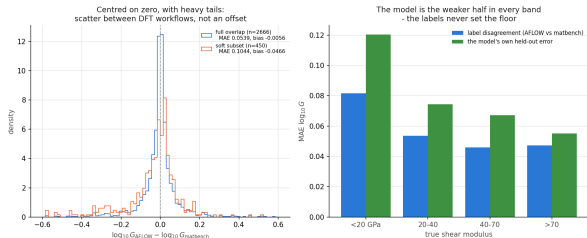
A decision tree on the composition-weighted mean of the same 92-dim element table – **no structure, no bonds** – on the identical split:

K MAE log ₁₀	tree	CGCNN
matbench	0.0868	0.0696
AFLOW	0.0592	0.1154

The cause is underfitting, checked directly.

Round 9's *training* error on AFLOW K is 0.103 – worse than the tree's *test* error of 0.059. A model that cannot fit its own training data is not out-generalised.

The label noise everyone quotes was the wrong number



A single constant – $|\Delta \log_{10} G| = 0.1085$ – was hard-coded into four scripts as *the* matbench/AFLOW label noise, and divided by a whole-test-set model error to argue the labels set the accuracy floor.

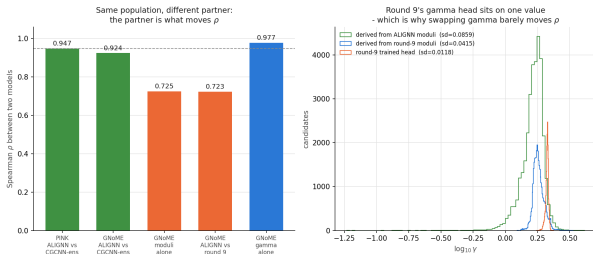
0.1085 is the *soft subset's* number.

Full overlap (n = 2,666): **0.0539**

Population-matched ratio: 0.69, not 1.39

Per stiffness band the *model's* error exceeds the label disagreement in **all four**.

Where a cross-model disagreement actually lives



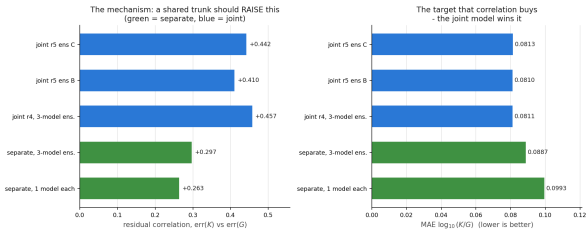
Agreement between ALIGNN and the CGCNN ensemble is $\rho = 0.947$ on the PINK set but only 0.723 against round 9 on GNoME. Two explanations were plausible: the exotic GNoME chemistry, or the partner model.

population cost (PINK $\rightarrow$ GNoME)	-0.023
partner cost (ens. $\rightarrow$ round 9)	-0.201

moduli alone	0.725
γ alone	0.977

Holding the partner fixed, GNoME's chemistry costs almost nothing. Swapping the partner costs nine times more – and it is entirely the moduli.

The joint model worked better than the record said



κ_L depends on K and G only through their *ratio*, and in log space a ratio is a difference – so independent errors **add**. A shared trunk should make them correlate instead.

	corr	MAE log ₁₀ (K/G)
separate, 1 each	+0.263	0.0993
separate, 3-ens	+0.297	0.0887
joint 3-ens	+0.457	0.0811

A published table had labelled the first two columns “separate” and “joint” – so it contained no joint model at all. The real correlation is 1.5× the ensemble’s.

The mechanism worked; it is paid for in per-modulus accuracy.

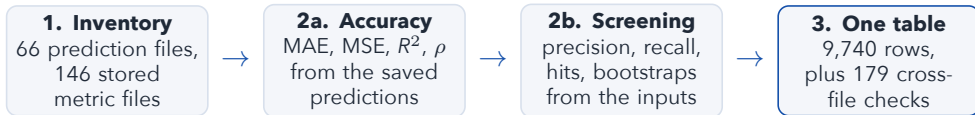
PART EIGHT

The audit

Every number in this deck, recomputed from its source

One file holds every metric of every model

Part 7 found numbers that were wrong, so every metric the project ever stored was **recomputed from scratch**. The formulas were re-written, not imported – an old bug cannot hide by being re-used.



column	what it holds
recomputed	the number rebuilt now
stored	the number in the old file (<code>stored_in</code>)
abs_diff	$ \text{recomputed} - \text{stored} $
rel_diff	$\text{abs_diff} \div \text{stored} $ – the gap as a fraction
status	the verdict, by the rule on the right
primary	True on the first row of each metric: filter on it to see every number once

The status rule

match: gap $\leq$ half a unit in the stored number's last digit (or 1 part in a million)

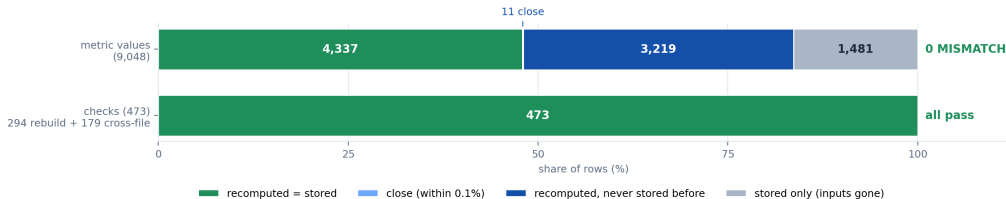
close: gap $\leq 0.1\%$

MISMATCH: anything larger

new: never saved before (e.g. R^2 for a run that kept only MAE)

stored only: carried over from its file, not rebuilt

The verdict: zero mismatches



4,337 rebuilt and identical

Every accuracy and screening number that had been reported, recomputed from its inputs and matched.

11 close, all below 0.1%

Models trained on a GPU and re-run on a CPU: the 4th-5th decimal moves. Same number, different arithmetic.

473 checks, all pass

294 rebuild checks + 179 cross-file ones, e.g. each Kaggle summary table equals its own JSON files.

9,048 distinct values, 0 MISMATCH. The 3,219 “new” values (R^2 , MSE, Spearman for runs that only kept MAE) mean every model now has the full set of metrics, not just the one that was quoted.

The moduli models, from that one file

model	bulk K		shear G		κ_L
	MAE	R^2	MAE	R^2	MAE
Random forest (composition only)	0.0868	0.841	0.1062	0.830	–
XGBoost (composition only)	0.0879	0.846	0.1128	0.820	–
CGCNN, 1 model (seed 42)	0.0696	0.882	0.0836	0.889	0.2065
CGCNN, 3-ensemble	0.0630	0.901	0.0781	0.899	0.1897
Joint CGCNN, 3-ensemble (r5 A)	0.0694	0.895	0.0836	0.887	0.1868
ALIGNN, 1 model (seed 42)	0.0549	0.917	0.0726	0.900	–
ALIGNN, 3-ensemble	0.0534	0.921	0.0708	0.904	0.1708

All on the same 1,648 matbench test crystals (`split_seed = 42`); MAE and R^2 in $\log_{10}$. The κ_L column scores the Slack chain on *predicted* moduli against the same chain on the *DFT* moduli (the in-house reference of slide 23), not against a phonon calculation. The trees never had a κ run; the joint model's per-modulus MAE is stored under its κ target and read from there.

ALIGNN's 3-ensemble is best in every column. Each graph network beats both composition-only trees on matbench – a ranking that Part 7 showed reverses on AFLOW for an underfit network.

Against measurement and phonon DFT

test set	model	measured as	value
PINK Table 1 45 measured κ_L	CGCNN ens., γ from Poisson	MAE $\log_{10}$	0.427
	CGCNN ens., Table 1's γ	MAE $\log_{10}$	0.226
	the PINK paper itself	MAE $\log_{10}$	0.227
PhoNIX phonon DFT 2,520 unseen	Poisson chain, max of 3 MLIP γ instead (451 sampled)	precision at $\kappa \leq 1$ precision gain	0.747 [0.704, 0.785] +0.013 [-0.069, +0.096]
PhoNIX, 5-fold 5,316 held out once	direct ALIGNN, structure $\rightarrow \kappa_L$	MAE $\log_{10}$	0.194
		R^2 / Spearman ρ	0.755 / 0.892

Measured κ_L (45)

With the paper's own γ we match the paper: 0.226 vs 0.227. With γ from Poisson's ratio the error doubles – γ , not the network, is the weak link.

Phonon DFT (2,520)

3 in 4 crystals the chain calls ≤ 1 really are. Swapping in a phonon γ from an MLIP adds +0.013 – its interval spans zero, so no measurable gain.

Direct ALIGNN

Trained straight on phonon-DFT κ_L , no Slack formula. Each crystal predicted once, while held out: typical miss $\times 1.56$ ($10^{0.194}$), rank $\rho = 0.89$.

Three traps the rebuild walked into

1. A tie at the 10% cut

mb-09580 and mb-00161 have identical labels (K 17, G 9 GPa) and share 164th place. The original step kept one and counted **108** hits; the rebuild kept the other and counted **109**.

Both are right. The check now accepts the whole range over every tie choice.

2. numpy changed how it sorts ties

Matbench moduli are whole GPa, so many crystals tie. numpy 2's fast (AVX2) sort orders ties differently from numpy 1.26: **41 MISMATCH + 23 close**, with no bug anywhere.

Forcing the old sort path (NPY_DISABLE_CPU_FEATURES=AVX2): **0**.

3. A label, not a number

Two γ models (gamma3, round 8) were listed as tested on the 835-crystal split. They were tested on the AFLOW γ set: **1,460** crystals, 1,022 for training.

The numbers were right; the description of them was not.

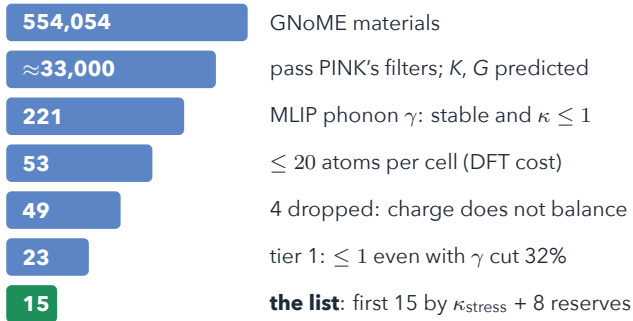
None of the three changes a conclusion – but two would have shown up as MISMATCH and sent someone hunting a bug that does not exist. An audit has to *explain* its differences, not only count them.

PART NINE

The DFT list

15 crystals handed over for a phonon-DFT κ_L

From 554,054 materials to 15 crystals



Ranked by safety, not by lowest value

The goal is a crystal whose DFT κ_L lands below 1 W/m/K, not one that hits a predicted number. So the ranking uses κ_{stress} : the *highest* of three moduli models, with γ cut by 32%.

Chemistry before physics

4 crystals need oxygen holes to balance charge – a known way for PBE to fake stability. Dropped. 3 others need rare oxidation states: moved to the reserves.

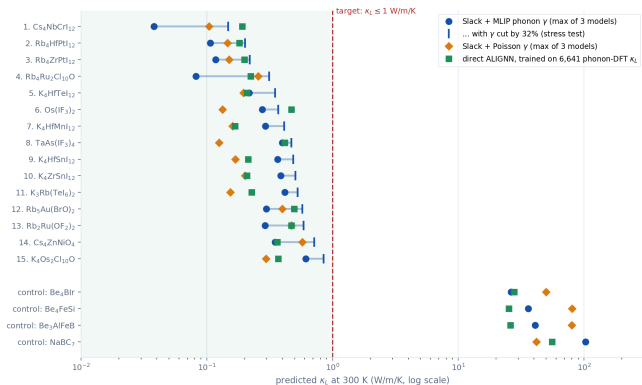
Every stage is a file in the repo (steps 83-91); the list is
`dft/final_shortlist_15/` **with CIFs, an index of every prediction, the**
4 controls and the 8 reserves.

The 15, as handed to the supervisor

#	formula	atoms	space group	supercell	spin-polarise	predicted κ_L , W/m/K, 300 K			
						max3	stress	Poisson	direct
1	Cs ₄ NbCrI ₁₂	18	P4/mmm (123)	2×2×1 (72)	yes, Cr(III)	0.04	0.15	0.10	0.19
2	Rb ₄ HfPtI ₁₂	18	P4/m (83)	2×2×1 (72)	no	0.11	0.20	0.15	0.18
3	Rb ₄ ZrPtI ₁₂	18	P4/m (83)	2×2×1 (72)	no	0.12	0.22	0.15	0.20
4	Rb ₄ Ru ₂ Cl ₁₀ O	17	I4/mmm (139)	2×2×1 (68)	check, Ru(IV)	0.08	0.31	0.26	0.22
5	K ₄ HfTeI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.22	0.35	0.20	0.21
6	Os(IF ₃) ₂	18	P $\bar{1}$ (2)	2×2×2 (144)	yes, Os(V) + I ₂ ⁺	0.28	0.37	0.13	0.47
7	K ₄ HfMnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	yes, Mn(IV)	0.29	0.41	0.16	0.17
8	TaAs(IF ₃) ₄	18	Cm (8)	2×2×2 (144)	check, I ₂ ⁺	0.40	0.47	0.12	0.42
9	K ₄ HfSnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.36	0.49	0.17	0.21
10	K ₄ ZrSnI ₁₂	18	P $\bar{1}$ (2)	2×2×1 (72)	no	0.39	0.51	0.20	0.21
11	K ₃ Rb(TeI ₆) ₂	18	P1 (1)	2×2×1 (72)	no	0.42	0.53	0.15	0.23
12	Rb ₅ Au(BrO) ₂	20	Pbam (55)	2×2×1 (80)	no	0.30	0.57	0.40	0.50
13	Rb ₂ Ru(OF ₂) ₂	18	Ama2 (40)	2×2×2 (144)	yes, Ru(VI)	0.29	0.59	0.47	0.47
14	Cs ₄ ZnNiO ₄	10	C2/m (12)	3×2×2 (120)	yes, Ni(II)	0.35	0.72	0.57	0.36
15	K ₄ Os ₂ Cl ₁₀ O	17	I4/mmm (139)	2×2×1 (68)	check, Os(IV)	0.61	0.85	0.30	0.37

max3: Slack, MLIP γ , highest of 3 moduli models. **stress:** max3 with γ cut 32% (the ranking value). **Poisson:** Slack with γ from each model's Poisson ratio, highest of 3. **direct:** ALIGNN trained on 6,641 phonon-DFT κ_L . Supercell: smallest with every vector ≥ 10 Å. 8 of 15 are one isolated-octahedron iodide family; a 2026 DFT study of 18 Cs₂BX₆ found every $\kappa_L < 1$ (Cao, arXiv:2602.06501). None of the 15 appears in any paper except the GNoME database.

Three routes, and four controls



Blue: Slack with the MLIP phonon γ (max3), the bar running to the stress value.

Amber: Slack with γ from Poisson's ratio – shares the moduli, not the γ .

Green: direct ALIGNN – shares neither.

Highest value of *any* route on the 15: **0.85**

Lowest on the 4 controls: **25.3**

Poisson route's stricter 0.54 cut:
14 of 15 ($\text{Cs}_4\text{ZnNiO}_4$ 0.57)

Direct route: **15 of 15** ≤ 0.50

Agreement is weaker evidence than it looks: the direct model also calls 44 of the 53 earlier candidates low. The controls are what make the run informative.

What the DFT run will settle

Reading the result

Controls first. The 4 Be/B crystals (predicted 25-103) must come out far above the 15, or the method is not separating low from high at all.

Then the 15. If the phonon-DFT calibration (slide 32) carried over – precision 0.75, about 0.85 at a cut as strict as this list's – **11-13 of 15** would land ≤ 1 , not 15 of 15. These are GNoME crystals no model has seen: an expectation, not a promise.

Then the routes. Each DFT value is scored against stress, Poisson and direct – all three are already in `index.csv`.

Setup notes sent with the list

Six slanted cells (#5, 7, 9, 10, 11, 13): standardise before relaxing.

Spin-polarise Cr, Mn, Ni, Os(V), Ru(VI); check #4, #8, #15.

Heavy I, Te, Pt, Os, Au, Hf, Ta: spin-orbit per group practice.

MLIP soft modes to confirm: at most -0.077 THz.

Short of time: #2/#3 and #9/#10 are Hf/Zr twins – run one of each.

Reserves: use $\text{Cs}_2\text{CuAgO}_2$ and $\text{Fe}_3\text{SbCl}_7\text{O}$ first. The direct model doubts reserves 3-5 (2.2-2.6 W/m/K).

So far every number is model vs model or model vs database. This run is the first test of the screen's picks against the real quantity.

PAPER FIGURES 9 AND 10

What we cannot reproduce

The paper's last two figures are **DFT phonon calculations** on $\text{Ag}_3\text{Te}_4\text{X}$ – primitive structures, phonon dispersions, then κ_L against temperature, cumulative κ_L , specific heat, mode group velocities, scattering phase space and scattering rates.

Producing them needs a full DFT plus finite-displacement phonon workflow with **third-order force constants**, on materials this project has never computed – days of DFT per material, and a different toolchain from anything here.

No stand-in is drawn for them. They are the same missing ground-truth gap the 15-crystal κ_L DFT list (Part 9) starts to close.

IN ONE SLIDE

What this work established

The pipeline finds a physical signal. Two architectures sharing only data and physics agree at $\rho = 0.947$ on κ_L .

ALIGNN earns its extra structure. Bond angles via the line graph improve both moduli – the only model change that survived re-scoring.

On equal footing we match the paper against measured κ_L : 0.226 against 0.227.

Most reported gains here are measurement artefacts: a circular metric, an over-confident interval, an underfit network flattered by comparison, a label-noise constant from the wrong population.

Every number is now reproducible: 9,048 metric values rebuilt from their inputs, 0 mismatches.

And the test is in motion: 15 crystals, 4 controls, 8 reserves handed over for phonon-DFT κ_L .